

Snippets

Arithmetic Snippets:

1. `int a= 10, b=15, c=0;`
`c= a++ + --a - b--`
`a=? b=? c=?`
2. `int a=10, b=20;`
`a -= a-- + --b;`
`a=? b=?`
3. `int a= 2, b=4;`
`a*= a-- * a/b;`
`a= ?`
4. `int a= 2, b=4;`
`a*= a-- * a%b`
5. `int a= 3, b=6;`
`double c= a/b + b/a;`
6. `System.out.println(2+3);`
7. `System.out.println("Sum="+2+3);`
8. `System.out.println("Sum"= + (2+3));`
9. `String s= "String";`
`int a=10;`
`int b=20;`
`s=s+a+b;`
`System.out.println(s);`
10. `System.out.println(1/2+ "\t"+ 1.0/2.0 + "\n" + "YO");`

Mathematical Snippets:

1. `double a= Math.pow(Math.pow(2,3),2);`
`a=?`
2. `double p= Math.max(Math.round(2.5), Math.ceil(2.2));`
`p=?`
3. `System.out.println(Math.pow(25, 1/2)+4);`
4. `System.out.println(Math.pow(25, 1.0/2.0)+4);`
5. Write the possible range of values for the expression `Math.random() * 10;`

hint: `Math.random()` returns a real value `x` where $0 \leq x < 1$
6. `System.out.println(Math.sqrt(1-10));`

6. Write the following expression in java:

$$x = \frac{|x-y|}{\sqrt{a^2-b^2}}$$

7. `System.out.println(Math.round(4.3));`

Conditional Snippets:

1. `double x= 2.5, y=2.5;`

`if(Math.floor(x)> y){`
`System.out.println(x+y);`
`}else{`
`System.out.println(x+""+y);`
`}`
2. `int x= 2>3 ? 5 : 4 >3 ? 3 : 6;`
`x=?`
3. find output if (i) choice= 'a' (ii) choice='x'

`switch(choice){`

`case 'b': System.out.println("No");`
`break;`
`case 'a': System.out.println("That is");`
`case 'c': System.out.println("Correct !");`
`break;`
`default: System.out.println("Wrong Input");`

`}`

4. Convert the if-else construct to ternary

```
if(x==4){
    v= v-0;
}else if( x<4){
    v= v+1;
}else{
    v=v
}
```

5. Convert the if else statement to switch case

```
if(x==3){
    System.out.println("React");
}else if(y==4){
    System.out.println("Express");
}else{
    System.out.println("Solidity");
}
```

Find the type of error (if any)

1. `int a= "Hello";`
2. `int x=10, y=5;`
`int c= (x*y)/(x%y);`
3. `int a= Math.pow(2,3);`
4. `System.out.print("Sum="(2*4));`

Conversion between loops (NO OUTPUT REQUIRED)

1. Convert to `while()` and `do while()`

```
int a=10;
for(int i=1; i< a/2; i++){
    a+=i;
}
```

2. Convert to `for` and `while()`

```
int k=0;
while(k<=50){
    System.out.println("Loop");
    k= k+5;
}
```

for loop snippets

1.

```
for(int m=5; m<=20; m+=5){
    if(m%3==0) break;
    if(m%5==0) System.out.println(m);
}
```
2.

```
for (int i = 1; i <= 10; i++) {
    if (i % 3 == 0) {
        continue;
    }
    System.out.print(i + " ");
}
```
3.

```
for (int i = 1; i <= 10; i++) {
    if (i * i > 30) {
        break;
    }
    System.out.print(i + " ");
}
```
4.

```
for (int i = 1; i <= 10; i++) {
    if (i % 2 == 0) {
        i++;
    }
    System.out.print(i + " ");
}
```

5.

```
for (int i = 10; i >= 1; i--) {
    if (i % 4 == 0) {
        continue; // Skip multiples of 4
    }
    if (i == 5) {
        break; // Exit the loop if i equals 5
    }
    System.out.print(i + " ");
}
```
6.

```
for (int i = 1; ; i++) {
    if (i % 7 == 0) {
        System.out.print("Lucky! ");
        break;
    }
    System.out.print(i + " ");
}
```

while loop snippets:

1.

```
int i = 1;
while (i <= 10) {
    if (i % 3 == 0) {
        i++;
        continue;
    }
    System.out.print(i + " ");
    i++;
}
```
2.

```
int i = 1;
while (true) {
    if (i * i > 20) {
        break;
    }
    System.out.print(i + " ");
    i++;
}
```
3.

```
int i = 1;
while (i <= 15) {
    if (i % 4 == 0) {
        i += 2;
        continue;
    }
    System.out.print(i + " ");
    i++;
}
```
4.

```
int i = 5;
while (true) {
    if (i % 7 == 0) {
        System.out.print("Found it: " + i);
        break;
    }
    i++;
}
```

```

5. int i = 10;
while (i > 0) {
    if (i % 2 == 0) {
        i -= 3;
        continue;
    }
    System.out.print(i + " ");
    i--;
}

```

do-while snippets

```

int i = 1;
do {
    if (i == 4) {
        i++;
        continue;
    }
    System.out.print(i + " ");
    i++;
} while (i <= 6);

```

```

2. int i = 1;
do {
    if (i == 5) {
        break;
    }
    System.out.print(i + " ");
    i++;
} while (i <= 10);

```

```

3. int i = 1;
do {
    if (i % 2 == 0) {
        i++;
        continue;
    }
    System.out.print(i + " ");
    i++;
} while (i <= 10);

```

```

4. int i = 1;
do {
    if (i == 3) {
        i++;
        continue;
    }
    if (i == 6) {
        break;
    }
    System.out.print(i + " ");
    i++;
} while (i <= 10);

```

nested loop snippets:

```

1. for (int i = 1; i <= 3; i++) {
    int j = 1;
    while (j <= 2) {
        System.out.print(i + " " + j + " ");
        j++;
    }
}

```

```

2. int i = 1;
while (i <= 2) {
    for (int j = 1; j <= 2; j++) {
        System.out.print(i + " " + j + " ");
    }
    i++;
}

```

```

3. for (int i = 1; i <= 2; i++) {
    int j = 1;
    do {
        System.out.print(i + " " + j + " ");
        j++;
    } while (j <= 2);
}

```

```

4. int i = 1;
while (i <= 3) {
    int j = 1;
    do {
        System.out.print(i + " " + j + " ");
        j++;
    } while (j <= 3);
    i++;
}

```

```

5. int i = 1;
while (i <= 2) {
    int j = 1;
    while (j <= 2) {
        System.out.print(i + " " + j + " ");
        j++;
    }
    i++;
}

```

```

6. outer: for(int i=2, k=3; i<=5; i++, k+=3){
    int j=1;
    while(j<=i){
        if(i==3)break outer;
        System.out.print(k+" ");
    }
    System.out.println();
}

```